

Algorithm Analysis & Complexity Theory

Sample Exam: Asymptotic Analysis

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Difficulty Levels: Foundation, Conceptual, Application, Advanced

Question 1 [Difficulty: Foundation]

Prove that $3n^2 + 5n + 2 = O(n^2)$. Use the formal definition of Big-O notation.

Solution:

To prove $f(n) = O(g(n))$, we need to show: $\exists c > 0, n_0 > 0$ such that $f(n) \leq c \cdot g(n)$ for all $n \geq n_0$

Let $f(n) = 3n^2 + 5n + 2$ and $g(n) = n^2$

For $n \geq 1$:

- $3n^2 \leq 3n^2$
- $5n \leq 5n^2$ (since $n \leq n^2$ for $n \geq 1$)
- $2 \leq 2n^2$ (since $1 \leq n^2$ for $n \geq 1$)

Therefore: $3n^2 + 5n + 2 \leq 3n^2 + 5n^2 + 2n^2 = 10n^2$

Choose $c = 10$ and $n_0 = 1$

Then $f(n) \leq 10 \cdot g(n)$ for all $n \geq 1$

$\therefore 3n^2 + 5n + 2 = O(n^2)$ ■

Question 2 [Difficulty: Conceptual]

Show that $n \log n \neq O(n)$. Prove by contradiction.

Solution:

Proof by contradiction:

Assume $n \log n = O(n)$

Then $\exists c > 0, n_0 > 0$ such that $n \log n \leq cn$ for all $n \geq n_0$

Dividing both sides by n (valid for $n > 0$):

$\log n \leq c$ for all $n \geq n_0$

But $\log n$ is unbounded as $n \rightarrow \infty$

For any constant c , we can find n_1 such that $\log n_1 > c$

(e.g., $n_1 = 2^{(c+1)}$)

This contradicts our assumption that $\log n \leq c$ for all $n \geq n_0$

Therefore, our assumption was false

$\therefore n \log n \neq O(n)$ ■